SLIs, SLOs and SLAs

measure what users feel, promise less than you aim for

Watch the 5-min video

In one line: an SLI measures user experience as good events ÷ valid events; an SLO is your internal target for it over a window (say 99.9% over 28 days); an SLA is a looser, external contract with penalties, so the gap between them gives you time to react before customers are owed money.

1 Healthy servers, unhappy users

The dashboard shows CPU at 40% and every host green. Meanwhile a third of checkouts fail, because a payment dependency returns errors that the host metrics never see. CPU, memory and "is the process up" describe machines, not users. Reliability is measured from the user's side, with **service level indicators**.

2 SLI, SLO, SLA

An **SLI** is a ratio: **good events ÷ valid events**, for example successful requests ÷ all requests, or requests served under 300 ms ÷ all requests. "Valid" excludes events you shouldn't count, such as health-check traffic or a client's own 400 errors. Common SLI types are **availability**, **latency**, **freshness** (how old the data is) and **correctness**.

An **SLO** is a target for an SLI over a **window**: "99.9% of requests succeed over a rolling 28 days". Engineering and product agree it together. Never aim for 100%: it is impossible, very expensive, and users can't tell 99.99% from 100% through their own Wi-Fi and mobile networks. An **SLA** is a contract that adds consequences, usually service credits, and is set **looser** than the SLO.

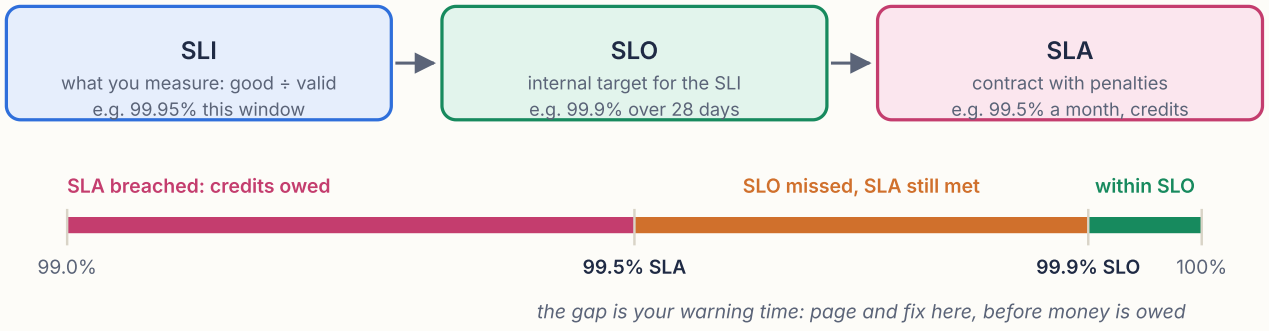


Figure 1. The SLI is the measurement, the SLO the internal target, the SLA the external promise. Missing the SLO is a warning; missing the SLA costs money.

3 Where and how you measure

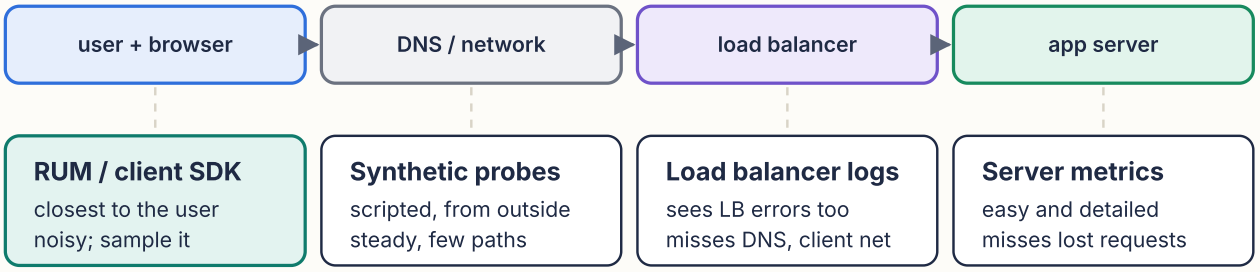


Figure 2. Each measurement point sees a different slice. Server logs can't count a request that never arrived.

- **Server-side** metrics are cheap and detailed but miss DNS failures, TLS errors and requests the load balancer rejected.
- **Load balancer** logs catch more, including 502/504s generated by the LB itself.
- **Client-side** measurement (real-user monitoring) and **synthetic probes** are closest to the real experience; most teams combine LB metrics for the SLO with probes for coverage.
- For latency, use **percentiles** (p99, p99.9), not averages: an average of 120 ms can hide 1% of users waiting 4 seconds. Better still, phrase it as a ratio: "fraction of requests under 300 ms".

4 Picking SLIs: a search API

SLI type	Good event	Valid events	Example SLO
Availability	search returns a non-5xx response	all search requests	99.9% / 28 days
Latency	answered in under 200 ms	all successful searches	99% / 28 days (i.e. p99 < 200 ms)
Freshness	new item searchable within 1 minute	all newly added items	99% / 28 days
Correctness	test query returns the expected top results	sampled golden queries	99.5% / 28 days

WORKED EXAMPLE: A CHECKOUT SLO

SLO: **99.9% of checkout requests succeed in under 500 ms, over 28 days**. The SLI counts a request as good only if it succeeds *and* is fast enough, so one SLI covers both availability and latency. With about 1,000,000 checkouts in the window, the SLO allows **1,000 bad ones** (0.1%). A checkout that succeeds in 2 seconds still counts as bad: to the user, a slow checkout feels broken. If the SLA promises 99.5%, the team has 4,000 more bad requests of warning before credits are owed.

COMMON MISTAKES

- Using CPU, memory or host uptime as SLIs: they describe machines, not users.
- Setting the SLA equal to the SLO, leaving no time to react before penalties.
- Averaging latency, which hides the slow tail that real users hit.
- Too many SLOs: pick 2–4 per user journey that you will actually act on (lesson 3).

5 Key terms

SLI: good events ÷ valid events, measured from the user's side

SLO: target for an SLI over a time window

SLA: external contract with penalties, looser than the SLO

Percentile (p99): value that 99% of requests are faster than

Rolling window: e.g. the last 28 days, recomputed continuously

Synthetic probe: scripted request sent regularly from outside

6 Interview questions

Q1. What is the difference between an SLO and an SLA? BASIC

An SLO is an internal reliability target for an SLI over a window; an SLA is an external contract with penalties such as credits, set looser than the SLO so you get warning time.

Q2. Why not set an SLO of 100%? BASIC

It's unachievable and very costly, it leaves no room for releases or maintenance, and users can't perceive the difference beyond their own network's reliability.

Q3. Pick SLIs for a search API. SCENARIO

Availability (non-5xx responses), latency (fraction under 200 ms, i.e. p99), freshness (new items searchable within a minute) and correctness (golden queries return expected top results), each as good ÷ valid events.

Q4. Servers look healthy but users complain. What do you change? SCENARIO

Define SLIs from the user's side: measure success and latency at the load balancer and with synthetic probes or RUM, so failures outside your servers are counted.

Q5. A checkout succeeds in 2 seconds. Is it a good event? TRICKY

Not if the SLI is "succeeds in under 500 ms": combining success and latency in one SLI counts slow successes as bad, because users experience them as failures.

Q6. Why are averages a poor latency SLI? TRICKY

A few very slow requests barely move the mean, so the tail that hurts users is hidden; percentiles or "fraction under threshold" expose it.

Next: SD-K-003 · Error Budgets backendroadmap.com